



Activities & Assessments

Programming and Artificial Intelligence

CHAPTER 2 | PART ONE | FIRST SEMESTER

Unit 2: Cybersecurity

3 lessons | Performance tasks | Home assignments | Weekly assessments

CHAPTER 2

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Unit 2: Cybersecurity

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HOW THIS WORKBOOK IS ORGANISED

1

Classroom Performances *(continuous formative assessment)*

Varied activities the teacher uses during the lesson to boost student interaction, check progress directly, and identify strengths and areas needing improvement.

2

Home Assignments *(mastery and self-application)*

Guided activities and exercises that help students learn independently, check their understanding, and apply what they have learned in new contexts.

3

Weekly Assessments *(performance measurement and follow-up)*

Assessment models carried out at the end of each week to track progress and measure the learning outcomes achieved during that week.

Part I of each period contains the classroom *Performance Tasks*; **Part II** contains the *Home Assignments*. Period 3 of every lesson holds the *Weekly Assessments* in three models (A, B and C). Each question has dotted lines for your written answer.

UNIT 2 | CYBERSECURITY

LESSON

1

Cryptographic Technologies and Authentication

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain how HTTPS communication protects the connection, and what role the TLS protocol plays in this.

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2. Distinguish between the role of common-key (symmetric) cryptography and public-key cryptography during the flow of an HTTPS connection.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following technologies is mainly used to encrypt session data quickly and efficiently once the handshake and security agreement are complete?
- A. Public-key cryptography. B. Common-key cryptography.
- C. Digital signature. D. Digital certificate.
2. What are the steps for establishing a secure connection and verifying the server's identity and domain name over HTTPS called?
- A. TLS handshake. B. Digital signature.
- C. Two-factor authentication. D. Common-key cryptography.

Part II | Home Assignments

Essay Questions:

1. What is meant by "two-factor authentication (2FA)", and how does using an independent second factor reduce the risk of unauthorized access?

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2. State the basic differences between two-factor authentication (2FA) and multi-factor authentication (MFA) in terms of the number and categories of factors used.

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Multiple-Choice Questions: (choose the correct answer)

- Which of the following represents two factors from two completely different categories (knowledge and possession) used in two-factor authentication for banking services?
 - Password + a second text password.
 - Password + a one-time password (OTP).
 - Fingerprint + face recognition.
 - Paper ID card + username only.
- What is meant by the term “multi-factor authentication (MFA)”?
 - Using a single, very strong and complex password.
 - Using two or more factors from different categories such as knowledge, possession, and biometrics.
 - Relying exclusively on a fingerprint alone to log in.
 - Using an email address together with a traditional password.

PERIOD 2

Week: Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

- What is the essential role of the digital certificate and public-key mechanisms in starting a secure TLS connection?

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- Briefly compare the function of a digital signature with the function of a digital certificate in security systems.

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Multiple-Choice Questions: (choose the correct answer)

- Which technology helps the browser verify the server's identity and that the certificate matches the domain name it is connecting to?
 - Common-key cryptography.
 - Digital signature.
 - Two-factor authentication.
 - Digital certificate.
- Which of the following technologies detects tampering and impersonation, and provides evidence that supports non-repudiation?
 - Common-key cryptography.
 - Digital signature.
 - TLS handshake.
 - Two-factor authentication.

Part II | Home Assignments**Essay Questions:**

1. Briefly explain how different security technologies (encryption, certificates, two-factor authentication) are combined when completing a purchase through an online shop.
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2. Why is a single security technology not enough to ensure the protection of services and communications over the Internet, and what is the practical benefit of multiple layers?
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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following technologies prevents the content of a connection from being read by third parties while data is transmitted across the network?

A. Encryption (common-key and public-key).	B. Digital signature.
C. Two-factor authentication (2FA).	D. Digital certificate.
2. Among the following options, which is an incorrect example of two-factor authentication because it relies on the same factor category (knowledge)?

A. Password + one-time password sent to the phone.	B. Password + fingerprint.
C. Password + a second secret password.	D. Password + a code from an authenticator app.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

1. Explain in detail how HTTPS communication uses different technologies to protect the connection, from the moment the handshake begins to the exchange of data.
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2. Distinguish precisely between the role of common-key cryptography and public-key cryptography, explaining why each is used at its stage of the HTTPS connection.
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3. Explain with a practical example how two-factor authentication is used when logging in to social-media accounts or banking services.
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4. Discuss the statement: "Secure communication and services over the Internet depend on combining encryption and authentication, not on a single technology." Show the role of each technology combined.

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Multiple-Choice Questions: (choose the correct answer)

- The HTTP protocol carried over a secured TLS connection, which provides data confidentiality and integrity during transmission, is known as:

A. HTTPS	B. FTP
C. DNS	D. SMTP
- Using two independent factors from two different categories to prove identity is known as:

A. Common-key cryptography.	B. Two-factor authentication (2FA).
C. Digital signature.	D. Digital certificate.
- A technology that detects tampering and impersonation and provides evidence that supports non-repudiation is:

A. Digital signature.	B. Public-key cryptography.
C. TLS handshake.	D. Cloud computing.
- Which of the following accurately represents the categories of factors in multi-factor authentication (MFA)?

A. Knowledge, possession, and biometrics.	B. Speed, efficiency, and cost.
C. Public key, private key, and password.	D. Wireless connection, local network, and cloud.

MODEL B

Essay Questions:

- Explain the basic steps that occur during the "TLS handshake", and how the browser verifies the server's certificate.

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- Explain why using public-key cryptography for all exchanged data would be slow, and how the TLS protocol addresses this problem using session keys.

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- Compare the different categories of authentication factors (knowledge, possession, biometrics), giving one example of each.

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4. Explain how using an independent second factor (such as an authenticator app or a text-message code) prevents an account from being compromised even if the main password leaks.

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Multiple-Choice Questions: (choose the correct answer)

- A pair of linked keys, made of a public key and a private key, used in cryptography to secure a connection, is:

A. Public-key cryptography.	B. Common-key cryptography.
C. Classical bit.	D. Quantum qubit.
- A service or system that helps the browser verify the server's identity and that the domain name matches the digital certificate is:

A. Digital signature.	B. Digital certificate.
C. Common-key cryptography.	D. Two-factor authentication.
- Which of the following is a correct example of the "possession" factor in two-factor authentication?

A. Knowing a secret question.	B. A one-time password sent to the phone, or a smart card.
C. The user's own fingerprint.	D. Recognizing facial features.
- Combinations of security technologies in online shops ensure that registration and payment are protected through:

A. Relying on a single technology for speed of execution.	B. Combining encryption, certificates, signatures, and two-factor authentication to reduce multiple risks.
C. Cancelling the use of passwords entirely.	D. Relying on common-key cryptography alone, without any authentication.

MODEL C

Essay Questions:

- Explain in detail how digital certificates and the chain of trust enable the browser to verify the authenticity and legitimacy of the server's website name.

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- Briefly discuss the digital signature, stating its properties and how it helps detect tampering and impersonation and provides non-repudiation evidence.

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- What are the security benefits of combining several security technologies (encryption, signatures, certificates, two-factor authentication) in the context of banking services and online shopping?

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4. Discuss freely why advanced security measures such as TLS and two-factor authentication do not mean protection is absolute or that a site's content is entirely trustworthy.
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Multiple-Choice Questions: (choose the correct answer)

1. The steps for establishing a secure connection and agreeing on shared secrets from which session keys are derived are called:
 - A. TLS handshake.
 - B. Common-key cryptography.
 - C. Digital signature.
 - D. Digital certificate.
2. Using two or more factors from different categories to prove identity (such as knowledge, possession, and biometrics) is called:
 - A. Multi-factor authentication (MFA).
 - B. Public-key cryptography.
 - C. TLS handshake.
 - D. Common-key cryptography.
3. Which of the following is an example of the "biometric" factor in authentication?
 - A. A text password.
 - B. Fingerprint or face recognition.
 - C. An ATM card.
 - D. An SMS verification code.
4. Secure communication and services over the Internet mainly rely on:
 - A. Combining encryption, authentication, certificates, and digital signatures.
 - B. One unique and absolute technology.
 - C. Using simple passwords only.
 - D. Eliminating the use of public and private keys.

UNIT 2 | CYBERSECURITY

LESSON

2

Network Security Design

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain what is meant by the concept of “defense in depth” in network security, and why providing multiple layers is safer than relying on a single means of protection.

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2. Explain the essential role of the firewall in network security, and give an example of how it controls traffic.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following is the accurate definition of the concept of “defense in depth” in network security?

A. Relying completely on a strong firewall at the network boundary.	B. Using multiple layers of security controls so that other layers reduce the risk if one layer fails.
C. Automatically trusting all devices located within the organization’s geographic area.	D. Encrypting all traffic exclusively when using public Wi-Fi networks.
2. What is the term for a system or program that monitors network traffic and allows or blocks it according to specific security rules?

A. Virtual private network.	B. Firewall.
C. Demilitarized zone.	D. Zero-trust approach.

Part II | Home Assignments

Essay Questions:

1. What is meant by the term “virtual private network (VPN)”? State two of its main uses.

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2. Why is placing a public web server in the “demilitarized zone (DMZ)” safer than placing it directly inside the organization’s internal network?

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Multiple-Choice Questions: (choose the correct answer)

1. What is the term for a private logical connection or network created over a public network such as the Internet, using encryption to protect data?
 - A. Firewall.
 - B. Virtual private network (VPN).
 - C. Demilitarized zone (DMZ).
 - D. Zero-trust approach.
2. Which of the following represents the most suitable and common use of a virtual private network (VPN) in modern work environments?
 - A. Connecting securely to the organization's network from home or while traveling and working remotely.
 - B. Protecting endpoint devices from viruses and removing them automatically without helper software.
 - C. Monitoring local network traffic and blocking spam email.
 - D. Providing platforms that let users publish and share content extremely quickly.

PERIOD 2

Week: Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

1. Explain what is meant by the “demilitarized zone (DMZ)” in network design, and how it protects internal infrastructure components.
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2. Briefly compare traditional security and the “zero-trust approach” in terms of how they treat internal network users.
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Multiple-Choice Questions: (choose the correct answer)

1. What is meant by the term “demilitarized zone (DMZ)” in network security structuring?
 - A. A fully open public network that lets attackers access sensitive databases.
 - B. A zone where servers exposed to the external network are placed separately from the organization's internal network.
 - C. Malicious software that monitors traffic and automatically changes firewall rules.
 - D. An encrypted virtual private line linking different branches over the public Internet.
2. What is the basic assumption on which the “zero-trust approach” relies to secure an organization's resources?
 - A. Treating every device or user inside the organization's network as automatically fully trusted.
 - B. Not automatically granting trust to a user or device merely because it is inside the network or owned by the organization, and checking identity and context.
 - C. Relying completely on the external security perimeter and not inspecting internal applications.
 - D. Encrypting only email messages, without regard to data-access permissions.

Part II | Home Assignments**Essay Questions:**

1. Explain how security assumptions about the traditional “perimeter” have changed with the spread of cloud computing and remote work.

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2. Give examples of the four security measures associated with the layers of defense in depth (network entry, communication path, server placement, endpoints).

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Multiple-Choice Questions: (choose the correct answer)

1. What is the concept that describes the traditional boundary, which assumes that inside the organization's network is completely safe and outside it is unsafe, and which cloud services and remote work have blurred?

A. Demilitarized zone.

B. Perimeter.

C. Edge computing.

D. Quantum superposition.
2. Which of the following “defense in depth” layers is directly related to installing antivirus software and updating the operating system?

A. Network entry.

B. Communication path.

C. Server placement.

D. Endpoints.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

1. Explain in detail how “defense in depth” protects an organization better than relying on a single means of protection, referring to what happens when one defense is breached.

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2. Explain the technical role of the “firewall” in monitoring network traffic, illustrating this with the protection of databases.

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3. State the detailed concept of the “virtual private network (VPN)” and show how it protects data transmitted over public Wi-Fi networks from eavesdropping.

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4. Compare in detail traditional security and the “zero-trust approach”, explaining why traditional designs alone are no longer sufficient today.

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Multiple-Choice Questions: (choose the correct answer)

- Which of the following devices or systems monitors network traffic and allows or blocks it according to specific security rules?

A. Firewall.	B. Demilitarized zone.
C. Virtual private network.	D. Edge computing.
- What is the zone called in which public-facing servers are isolated from the internal network to reduce the impact of attacks?

A. Perimeter.	B. Demilitarized zone (DMZ).
C. Communication path.	D. Endpoints.
- What is the term for a security approach that does not automatically grant trust to a user or device merely because it is inside the network?

A. Firewall.	B. Zero-trust approach.
C. Defense in depth.	D. Virtual private network.
- Which of the following represents the main use of a virtual private network (VPN) for connecting offices or employees?

A. Encrypting the connection and protecting data over untrusted networks or during remote work.	B. Automatically removing viruses found on the hard disk.
C. Speeding up public Internet connections without needing passwords.	D. Providing a graphical interface for managing large databases.

MODEL B

Essay Questions:

1. Explain how the “demilitarized zone (DMZ)” protects an organization’s internal network when one of its public web servers is exposed to an external attack.

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2. Discuss the four layers of “defense in depth”, stating the security measure associated with each layer from network entry to endpoints.

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3. Explain how applying the “zero-trust approach” contributes to verifying every access based on identity, permissions, and context.

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4. Discuss why the effectiveness of the traditional “perimeter” has declined in modern work environments that rely on cloud computing and remote work.

Multiple-Choice Questions: (choose the correct answer)

- The technology that creates an encrypted virtual private line over the Internet for secure connection from a remote location is:

A. Firewall.	B. Virtual private network (VPN).
C. Demilitarized zone.	D. Zero-trust approach.
- The defense-in-depth layer concerned with installing antivirus software and keeping the operating system updated is:

A. Network entry.	B. Communication path.
C. Server placement.	D. Endpoints.
- The traditional boundary that assumes the inside of the organization’s network is completely safe and the outside is dangerous is known as:

A. Perimeter.	B. Demilitarized zone.
C. Edge computing.	D. Quantum superposition.
- A system or program that monitors network traffic and allows or blocks it according to specific security rules is:

A. Firewall.	B. Virtual private network.
C. Edge computing.	D. Qubit.

MODEL C

Essay Questions:

- Discuss the security importance of using a “virtual private network (VPN)” when connecting to untrusted networks, and how it protects connection content from eavesdropping.
- Explain how the firewall, virtual private network, demilitarized zone, and endpoint controls integrate to form a complete “defense in depth” strategy.
- Show how the “zero-trust approach” handles identity, context, and permissions compared with traditional security, which trusts everything inside the perimeter.

4. Explain why placing an email server or web server in the “demilitarized zone (DMZ)” is an essential protective measure for the safety of sensitive internal databases.

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Multiple-Choice Questions: (choose the correct answer)

1. A zone where servers exposed to the external network are placed separately from the internal network is called:

A. Firewall.	B. Demilitarized zone (DMZ).
C. Virtual private network.	D. Endpoints.
2. A security approach that does not automatically grant trust to a user or device merely because it is inside the network or owned by the organization is:

A. Defense in depth.	B. Zero-trust approach.
C. Perimeter.	D. Firewall.
3. Which of the following represents the first layer (network entry) in the defense-in-depth strategy?

A. Encrypting the connection using a virtual private network.	B. Preventing unauthorized connections using a firewall.
C. Separating public-facing servers using a demilitarized zone.	D. Installing antivirus software on endpoint devices.
4. A private logical connection or network over a public network such as the Internet, using encryption and tunneling to protect data, is:

A. Demilitarized zone.	B. Virtual private network (VPN).
C. Zero-trust approach.	D. Firewall.

UNIT 2 | CYBERSECURITY

LESSON

3

Incident Response and Risk Management

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. What is meant by the term “security incident”, and what are its most typical examples?

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2. Explain the six stages of incident response, stating the essential purpose of the “preparation” and “detection” stages.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following is a typical example of a security incident that affects the availability of a service?

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|---|--|
| A. Personal or confidential information leaking to the outside. | B. Modifying data within a system without authorization. |
| C. Service outage due to attacks on the server. | D. Establishing response procedures and structures in advance. |

2. What is the first stage of the incident-response model, which aims to establish response procedures and structures in advance, in readiness for incidents?

- | | |
|-----------------|-----------------|
| A. Detection. | B. Preparation. |
| C. Containment. | D. Recovery. |

Part II | Home Assignments

Essay Questions:

1. State the purpose of the “containment” and “eradication (removing the threat)” stages in the incident-response model.

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2. How is the risk score estimated in the lesson’s simplified model? State its formula and its two basic factors.

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Multiple-Choice Questions: (choose the correct answer)

1. What is the term for the action of isolating the affected part or limiting its impact in a suitable way to prevent the problem from spreading?

A. Preparation.	B. Detection.
C. Containment.	D. Recovery.
2. How is the risk score calculated in the simplified matrix?

A. Impact + likelihood.	B. Impact × likelihood.
C. Impact – likelihood.	D. Impact ÷ likelihood.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

1. Explain what is meant by the “recovery” and “lessons learned and improvement” stages in the incident-response model.
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2. Why must the “containment” stage be carried out before the “eradication (removing the threat)” stage? What might happen if this order were reversed?
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Multiple-Choice Questions: (choose the correct answer)

1. Which stage involves returning systems and services to a safe and normal operating state, and may include restoring from a clean backup?

A. Preparation.	B. Containment.
C. Recovery.	D. Lessons learned and improvement.
2. Which stage concludes the six stages of incident response and involves analyzing what happened and documenting lessons to improve controls?

A. Detection.	B. Eradication.
C. Recovery.	D. Lessons learned and improvement.

Part II | Home Assignments**Essay Questions:**

1. Explain how risks are handled according to the “risk assessment matrix”, and which category represents the highest treatment priority.
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2. Give examples of overlap or repetition that may occur between the incident-response stages in real practice.

Multiple-Choice Questions: (choose the correct answer)

1. According to the risk assessment matrix, which category is considered the highest priority and must be addressed first?
 - A. Low impact and low likelihood.
 - B. Medium impact and medium likelihood.
 - C. High impact and high likelihood.
 - D. Low impact and high likelihood.
2. What is meant by the term “impact” in the context of risk assessment?
 - A. How likely the event is to occur.
 - B. How much damage occurs if the risk materializes.
 - C. The speed of response to the security incident.
 - D. The number of times the incident has occurred.

PERIOD 3

Weekly Assessments

Week:

Date:/...../.....

MODEL A

Essay Questions:

1. Define the security incident in detail and give three of its typical examples with a brief description of each.

2. State the six stages of incident response in sequential order, with a brief explanation of the preparation and containment stages.

- 3.** Explain how the risk assessment matrix is used to set treatment priorities based on impact and likelihood.

4. Discuss why it is important to carry out the six response stages in order, and what kind of overlap or repetition may occur between them during actual treatment.

Multiple-Choice Questions: (choose the correct answer)

1. An event that affects the confidentiality, integrity, or availability of information, or that violates a security policy, is called:
 - A. A security incident.
 - B. A risk assessment.
 - C. Threat containment.
 - D. A lesson learned.

2. The stage used to limit the spread of damage and isolate the affected part is called:
 - A. Preparation.
 - B. Containment.
 - C. Recovery.
 - D. Eradication.
3. The risks that represent the highest priority in the risk assessment matrix have:
 - A. Low impact and rapid containment.
 - B. High impact and high likelihood.
 - C. Medium impact and zero likelihood.
 - D. Small impact and low likelihood.
4. What does the “eradication” stage mean in the incident-response model?
 - A. Establishing response procedures in advance.
 - B. Removing the threat and its causes, such as deleting malware.
 - C. Restoring systems from a backup.
 - D. Documenting lessons learned.

MODEL B

Essay Questions:

1. Explain in detail the purpose of the “detection” and “eradication (removing the threat)” stages within the six stages of incident response.

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2. Explain the concepts of “impact” and “likelihood” in the context of risk assessment and how they contribute to calculating the risk score.

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3. Discuss the importance of the “lessons learned and improvement” stage and how it helps prevent security incidents from recurring in the future.

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4. Compare the three types of security incidents mentioned in the lesson (information leak, data tampering, service outage) in terms of description.

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Multiple-Choice Questions: (choose the correct answer)

1. Modifying data within a system without authorization is a typical example of:
 - A. Service outage.
 - B. Information leak.
 - C. Data tampering.
 - D. Advance preparation.
2. The stage concerned with returning systems and services to a safe and normal operating state, including restoring from a backup, is:
 - A. Recovery.
 - B. Containment.
 - C. Preparation.
 - D. Detection.

3. The risk score in the simplified model is calculated as follows:
- A. $\text{Impact} \div \text{likelihood}$.
 - B. $\text{Impact} + \text{likelihood}$.
 - C. $\text{Impact} - \text{likelihood}$.
 - D. $\text{Impact} \times \text{likelihood}$.
4. What is meant by the term “likelihood” in risk assessment?
- A. The expected size of the damage if the event occurs.
 - B. How likely the event or risk is to occur.
 - C. The speed of executing the response stages.
 - D. The number of devices affected by the attack.

MODEL C

Essay Questions:

1. Explain in detail how the “recovery” and “lessons learned and improvement” stages are carried out when dealing with a security incident such as a ransomware attack.
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2. Discuss the relationship between the “containment” and “eradication” stages, and why containment must precede removing the cause.
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3. State the six stages of incident response in sequence, and show how overlaps or returns to earlier stages can occur during an actual response.
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4. Explain how organizations use the risk assessment matrix to classify risks and direct efforts toward effective initial treatment.
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Multiple-Choice Questions: (choose the correct answer)

1. The leak of personal or confidential information to the outside represents a security incident of the type:
 - A. Information leak.
 - B. Service outage.
 - C. Data tampering.
 - D. Safe recovery.
2. Analyzing what happened, documenting the lessons, and improving controls to reduce the likelihood of recurrence takes place in the stage of:
 - A. Preparation.
 - B. Containment.
 - C. Eradication.
 - D. Lessons learned and improvement.
3. Establishing response procedures and structures in advance, in readiness for incidents, is the stage of:
 - A. Preparation.
 - B. Detection.
 - C. Containment.
 - D. Recovery.

4. What is meant by the term “security incident”?

- A. A situation in which actual damage occurs due to an information security problem.
- B. A technology that encrypts communication data.
- C. A device that monitors network traffic.
- D. A method of authenticating a user’s identity.